

Stress-Based Fatigue Analysis - Final UI/UX Review

(2026-07-12)

Scope: bounded production-efficiency review and implementation pass focused on current product intent (no speculative redesign).

Final release/UIUX verdict

Acceptable for release now as a screening-oriented engineering estimator, with no UI/UX blockers remaining from this scoped pass.

Reason: high-impact friction points were addressed (safer reliability default, tighter input flow copy, clearer preview labeling, and better action guidance for reliability-format errors).

What was improved in this pass

- Reliability flow hardened: Weibull estimation is opt-in by default and no longer auto-seeded with demo-like data.
- Added explicit demo data loader action for quick UI checks without implying production-grade statistical confidence.
- Reduced repetitive micro-copy in primary input flow; consolidated symbol guidance into a compact, scannable note.
- Improved reliability error remediation guidance with concrete valid examples (e.g., "25000, fail").
- Reframed materials-selection scaffold as a developer preview while preserving functionality.
- CLI operator efficiency improved via concise final run summary (sample count, successes, failures).

Remaining TODO backlog (optional, non-blocking)

- **TODO (optional):** Add lightweight UI regression checks for key defaults (unit mode, Weibull toggle default off, empty reliability input).
- **TODO (optional):** Add a compact/advanced input presentation toggle to further reduce data-entry time for repeat expert users.
- **TODO (optional):** Introduce copy-level consistency review across all captions/help text to keep terminology identical between app panels.
- **TODO (optional):** Add short in-app examples for common reliability datasets (minimum valid rows) without enabling them by default.
- **TODO (optional):** Add screenshot-based release checklist for critical pages to catch accidental layout regressions before deploy.

Practical production notes

The current UX remains intentionally engineer-forward and mathematically explicit. The implemented changes optimize speed and decision safety while preserving the app's established workflow and domain language.

This final review intentionally avoids broad redesign proposals. The backlog above is optional and can be scheduled incrementally with low implementation risk.

Prepared by: UI/UX production-efficiency pass (Copilot session)